Process Characterization Plan

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCP-010

Process Development / Manufacturing Science & Technology (MSAT)

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

A280, protein concentration measured by ultraviolet absorbance. ALCOA+, the data integrity attributes (attributable, legible, contemporaneous, original, accurate, and complete, consistent, enduring and available). AMV, analytical method validation. BLA, Biologics License Application. CQA, critical quality attribute. DS, drug substance. DV, diavolume. HMW, high molecular weight. icIEF, imaged capillary isoelectric focusing. MSAT, Manufacturing Science and Technology. NOR, normal operating range. PAR, proven acceptable range. PD, Process Development. QA, Quality Assurance. SDM, scale-down model. SEC, size exclusion chromatography. SOP, standard operating procedure. TFF, tangential flow filtration. TMP, transmembrane pressure. UF/DF, ultrafiltration and diafiltration.

1 Purpose and scope

This plan defines the process characterization studies for the ultrafiltration and diafiltration step of the A-Mab drug substance process, which is Step 10 and the last step of the drug substance train. The process is being transferred from development in Cambridge to the commercial drug substance facility in Grafton, and the studies described here form part of the Stage 1 process design record that supports the transfer and the Biologics License Application (U.S. Food and Drug Administration 2011; International Council for Harmonisation 2008). The commercial process is operated at a production bioreactor working volume of 15,000 L and the step delivers drug substance at 75 g/L.

The plan covers the three process parameters listed in §6.1, the scale-down model on which they will be studied, the analytical methods and sampling, the acceptance and decision criteria, and the analysis by which a proven acceptable range will be assigned to each parameter. It applies to the drug substance process only. Drug product manufacture and the development of the formulation itself are covered by the drug product programme and are not in scope here. The composition of the formulation buffer is fixed by that programme and is treated in these studies as a constant.

The step is not where the quality attributes of A-Mab are formed. Concentration and buffer exchange neither create nor remove the glycan and impurity attributes. What the step can add to the size variants and to the charge variants is described in §4.1, and the studies measure that increment against the material Step 9 delivers. No viral clearance is claimed for this step. Clearance is credited to the low pH hold, the anion exchange step and the small virus retentive filtration step, and the cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a).

2 Objectives

The studies described in this plan will meet five objectives.

(i) Qualify a bench scale TFF model of the commercial skid and demonstrate that it represents the commercial step for the responses this plan measures.

(ii) Determine the effect of the number of diavolumes, the transmembrane pressure and the final DS concentration on the aggregate content and the charge variant distribution of the drug substance, over the characterization ranges given in §6.1.

(iii) Establish the number of diavolumes at which the retentate reaches the formulation buffer, so that the lower edge of the operating range rests on data for A-Mab rather than on platform practice alone.

(iv) Establish a PAR for each of the three parameters against the drug substance acceptance criteria in §4.2, by the analysis described in §8.

(v) Provide the evidence on which the three parameters will be classified under SOP-4001, and deliver that classification to the control strategy through PCR-010 and PCMR-001.

3 Related documents

This plan sits under the transfer plan and the master plan, takes its study scope from the pre-characterization risk assessment, and is executed and reported by PCR-010.

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-010	Process Characterization Report (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and validated methods that apply to the studies are listed in Table 3.2. Operation of the step follows SOP-2013, qualification of the scale-down model follows SOP-1001, and the classification of the parameters after the studies follows SOP-4001.

Table 3.2: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The UF/DF step concentrates the antibody and exchanges the process buffer for the formulation buffer. Both operations run across a tangential flow membrane whose pores pass buffer salts and retain the antibody. Ultrafiltration removes permeate at constant membrane area and raises the antibody concentration in the retentate. Diafiltration adds formulation buffer at the rate at which permeate leaves, so the volume is constant and the permeable species are washed out. Each diavolume removes a fixed fraction of the permeable species that remain, and the residual process buffer components therefore fall exponentially with the number of diavolumes. The step ends with a second concentration to the DS target of 75 g/L and a buffer flush that recovers the retentate.

The principal route from the step to the product is mechanical. The antibody the membrane retains accumulates at the membrane wall in a polarisation layer, where its concentration is higher than in the bulk retentate, and the whole retentate passes repeatedly through the recirculation pump. Aggregate can form in both places. The risk rises with the final concentration, with the transmembrane pressure that raises the antibody concentration at the wall, and with the number of passes through the pump. Within the normal operating ranges the exposure is short and the effect is expected to be small.

The step also holds the antibody in the formulation buffer for the length of the operation, and the chemical routes that run in solution continue during that hold. Deamidation of asparagine residues is the one that bears on a measured attribute, and it raises the acidic charge variants at a rate that rises with pH and with the length of the hold. The number of diavolumes sets how long the hold lasts, which is why the diavolume series is assayed for charge variants as well as for aggregate.

The step has been operated on the Novacyte humanized IgG1 platform for X-Mab, Y-Mab and Z-Mab, with the same membrane chemistry and the same mode of operation as is proposed for A-Mab (CMC Biotech Working Group 2009). The mechanisms above do not depend on the sequence of the antibody, so the platform experience applies directly to the qualitative behaviour of the step. The viscosity of the retentate does depend on the product, and it rises steeply as the antibody concentration approaches the DS target. A higher viscosity raises the pressure drop along the feed channel and the shear the antibody sees near the end of the concentration. For this reason the final DS concentration is studied here rather than carried over from the platform.

4.2 Quality attributes in scope

Two drug substance attributes are measured across the step, and they are listed in Table 4.1 with their acceptance criteria and their criticality from the A-Mab attribute assessment (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). The Tool #1 column carries the score of the first criticality tool of that assessment, which is the impact of the attribute on safety and efficacy multiplied by the uncertainty in that impact. The criticality level is assigned separately and is not read off that score. An attribute whose impact is well established carries a low uncertainty factor, so a low score can sit against a high level.

Table 4.1: Quality attributes measured across the step, with drug substance acceptance criteria.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Aggregates are the attribute most at risk at this step, by the mechanical route described in §4.1, and they carry a criticality of H. The acceptance criterion is 0–5 % HMW (SEC). Acidic charge variants carry a criticality of VL and a criterion of 18–40 % (CEX/icIEF). They are measured because deamidation of asparagine residues continues while the antibody is held in the formulation buffer, and the rate of deamidation rises with pH and with the length of the hold. A rise in acidic species across the step would show that the hold has altered the molecule. Both attributes arrive with the load. Aggregate and the acidic charge variants are formed in the production bioreactor, and aggregate is reduced again at the cation exchange step. The study measures what this step adds to each, against what Step 9 delivers.

The remaining attributes of the register are not measured across the step. The glycan attributes and high mannose are formed in the production bioreactor and are not modified by downstream processing, so no measurement at this step would inform them. Host cell protein, DNA and leached Protein A are cleared upstream. The membrane retains these impurities with the antibody, so the impurity to product ratio that Step 9 delivers carries through the step unchanged and the step neither clears nor concentrates them relative to the antibody. Viral clearance is not measured here, for the reason given in §1.

4.3 Risk-based prioritization of parameters

RA-001 ranked the parameters of every step of the drug substance process for their potential impact on the CQAs and for their potential to interact with other parameters (International Council for Harmonisation 2023b). The study type of each parameter was assigned from that ranking. For this step the assessment placed all 3 parameters below the threshold that warrants a multivariate design, because the step sets no CQA of its own and the three parameters reach the product by different routes. More diavolumes wash more of the permeable process buffer components out of the retentate, and they add time on the skid and passes through the pump. Higher transmembrane pressure raises the permeate flux until the polarisation layer limits it, and beyond that point it raises the antibody concentration at the membrane wall without raising throughput. A higher final DS concentration raises the bulk concentration of the retentate, and with it the viscosity, the pressure drop along the feed channel and the antibody concentration at the membrane wall.

Both the transmembrane pressure and the final DS concentration raise the antibody concentration at the membrane wall, the first by driving more permeate through the layer and the second by raising the bulk concentration that supplies it. An interaction between them is therefore possible. A study that varies one factor at a time cannot resolve such an interaction. What the design does bound is the effect of each parameter at the edge of its characterization range with the other parameters at the values the commercial step will hold. Where a measured attribute approaches its acceptance criterion at an edge, the study should be extended to the combination of edges before a range is claimed, and the extension should be agreed with Statistics and QA before it is executed.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run on the bench scale TFF skid EQ-TFF-142, which is registered as the ultrafiltration / diafiltration TFF skid and is held under calibration and change control. Its calibration status at execution will be recorded in the study record.

Two quantities decide what the antibody experiences in the step, and the model holds both at their commercial values. The membrane loading in grams of antibody per square metre fixes the bulk concentration the step reaches and the time the antibody spends on the skid. The crossflow per unit membrane area sweeps retained antibody from the membrane wall back into the bulk, so a higher crossflow lowers the antibody concentration in the polarisation layer. The membrane chemistry, the nominal molecular weight cut-off and the feed channel geometry are the same as on the commercial skid. Transmembrane pressure and the number of diavolumes are scale independent and are set at the values under study.

One quantity cannot be matched. The ratio of pump displacement to system hold up volume is larger at bench scale, so the antibody passes through the pump more often per gram processed than it does at commercial scale. The bench model is therefore conservative for aggregate formation by shear.

Qualification follows SOP-1001. Triplicate runs at the set-point condition will be compared with the pilot and commercial record on step yield, on permeate flux at the set-point transmembrane pressure, on the final concentration achieved, and on the change in aggregate content across the step. The acceptance criteria for the comparison are given in §7.1. No characterization run will be executed until the model has been accepted. If the comparison does not meet its criteria, the cause should be investigated and the model modified before the exercise is repeated.

5.2 Operation

The step receives the filtrate pool from the small virus retentive filtration step. The pool is concentrated by ultrafiltration, diafiltered at constant volume against formulation buffer for the planned number of diavolumes, concentrated a second time to the final target, and recovered with a buffer flush. Operation follows SOP-2013. At the set-point condition the step runs 8 DV at a transmembrane pressure of 12 psi to a final concentration of 75 g/L.

All runs in the study will be loaded from a single Step 9 pool, so that the variability of the load does not confound the comparison between levels of a parameter. The bridging rule that applies if one pool is not sufficient is given in §12.

5.3 Analytical methods

The validated methods are listed in Table 3.2. Aggregate is measured by SEC under AMV-3011 and reported as the percentage of high molecular weight species. Protein concentration is measured by A280 under AMV-3019, and it is the measurement on which the mass balance and the final concentration decision rest. Charge variants are measured by icIEF under AMV-3013 and reported as the percentage of acidic species. The completeness of the buffer exchange is followed by the conductivity and the pH of the retentate. These are in-process measurements read from the skid instrumentation and are not validated assays, so they are used to characterize the exchange curve and not to release a run.

The aggregate content of A-Mab drug substance is low, and a change across this step may be of the same size as the repeatability of the SEC method. Analytical Development will report the method variability that applies over the range in which the samples fall. A measured change is judged against the pure error estimate described in §5.5 rather than against the nominal precision of the method.

5.4 Sampling plan

Samples will be taken at the points listed in Table 5.1. The acceptance criteria of §7.2 apply to the final retentate. The other samples support the mass balance, the buffer exchange curve and the confirmation that the membrane retains the antibody.

Table 5.1: Sampling plan for each characterization run.

Sample	Point in the step	Tests	Purpose
Load pool	Step 9 filtrate, before ultrafiltration	A280, SEC, icIEF	Fixes the input to every run in the study
Pre-diafiltration retentate	End of the first concentration	A280	Confirms the concentration at which diafiltration starts
Diafiltration series	After each diavolume	Conductivity, pH	Builds the buffer exchange curve
Final retentate	After the final concentration and recovery flush	A280, SEC, icIEF	The drug substance sample the acceptance criteria apply to
Permeate	Composite over the run	A280	Confirms that the membrane retains the antibody

Every sample will be assayed within the stability period established for the matrix, and the load pool sample will be assayed once for the whole study rather than once per run. Where a run is repeated, the repeat carries its own sample set.

5.5 Statistical methods

For each parameter and each response, the measured values will be regressed on the level of the parameter over the characterization range, and the slope will be tested against zero at $\alpha = 0.05$. The set-point condition is shared by the three series and will be run 3 times. Those replicates give the pure error estimate of the combined process and assay variability, which is the band against which a change across a range is read.

Statistical significance and practical significance will be reported separately. A slope that is significant at $\alpha = 0.05$ but predicts a change across the range that is small against the acceptance criterion will be described in those terms, and the criterion rather than the p-value decides whether the range is acceptable. No response surface model will be fitted for this step, because the design carries no factorial structure and no interaction or curvature term can be estimated from it.

The buffer exchange will be analysed separately. The conductivity of the retentate will be plotted against the number of diavolumes, and the exponential decay expected from §4.1 will be fitted to the series. The analysis reports the number of diavolumes at which the conductivity of the retentate has reached the plateau set by the formulation buffer and no further change is measured over the following diavolume.

6 Study design

6.1 Factors, ranges and study type

The parameters to be studied, their set-points, the ranges to be characterized and the normal operating ranges are given in Table 6.1. All three are studied one at a time, for the reasons given in §4.3.

Table 6.1: Parameters, characterization ranges and study type for the step.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Number of diavolumes	DV	8	7–10	7–9	univariate
Transmembrane pressure	psi	12	10–15	10–14	univariate
Final DS concentration	g/L	75	65–85	70–80	univariate

Each characterization range is wider than the corresponding NOR, and it is extended in the direction in which the mechanism of §4.1 places the risk. The number of diavolumes is studied from 7 to 10 DV against a NOR of 7 to 9 DV. The range is extended above the NOR because more diavolumes mean more time on the skid and more passes through the pump. Its lower edge coincides with the lower edge of the NOR, which is the lowest count the commercial step will run, and the buffer exchange curve characterizes the step at that value. Transmembrane pressure is studied from 10 to 15 psi against a NOR of 10 to 14 psi, and the range is extended above the NOR because pressure above the flux limit compresses the polarisation layer without raising throughput.

The final DS concentration is studied from 65 to 85 g/L against a NOR of 70 to 80 g/L, and the range is extended on both sides. An excursion above the target carries the aggregation risk described in §4.1. An excursion below the target leaves the drug substance below the concentration the drug product programme expects, and it has to be dispositioned as readily as an excursion above it.

The step will not be characterized outside these ranges, and no range wider than the range studied will be proposed in PCR-010.

6.2 Univariate assessment

Each parameter will be run at 3 levels, which are the two edges of its characterization range and the set-point, with the other two parameters held at their set-points. The set-point condition is common to all three series, so the study consists of 6 edge runs and 3 replicates of the set-point condition, which is 9 runs in total. The planned levels are listed in §14.

The order of execution will be randomized across the whole study, subject to the constraint that all runs draw from the load pool of §5.2. Each run will be sampled as described in §5.4 and the final retentate will be assayed for the panel of §5.3. The diavolume series carries the additional in-process sampling that builds the buffer exchange curve.

The design gives no estimate of an interaction between parameters.

7 Acceptance and decision criteria

7.1 Scale-down model

The model will be accepted when the mean step yield, the permeate flux and the change in aggregate content across the step do not differ from the pilot and commercial record at $\alpha = 0.05$, and when the final concentration achieved is within the reported variability of the A280 method. A difference that is statistically significant will be assessed for its size against the acceptance criteria of §7.2 before the model is rejected, because a small and well understood offset may still support a conservative model. Any such judgement will be documented in the qualification record and approved by Statistics and QA.

7.2 Product quality

The acceptance criteria applied to the final retentate are the drug substance criteria of Table 4.1. Aggregate must fall within 0–5 % HMW (SEC) and acidic charge variants within 18–40 % (CEX/icIEF). These criteria apply directly at this step and are not reduced to an in-process limit. The step is the last of the drug substance process, and no step downstream of it clears or modifies either attribute, so there is no clearance chain through which a tighter in-process limit could be calculated backwards.

The buffer exchange is accepted when the conductivity and the pH of the retentate have reached the plateau of the formulation buffer, by the analysis of §5.5, at or below the lower edge of the diavolume NOR of 7 DV.

7.3 Decision rules

A level of a parameter is acceptable when every attribute measured on the final retentate of that run meets its criterion. A parameter passes over its characterization range when every level in the range is acceptable and the change measured across the range is small against the criterion.

Where a level fails, the range is truncated at the last acceptable level and the failing level is reported as an edge of failure. Where a failure occurs at a level inside the NOR, execution will stop and the operating range will be reviewed with the receiving site and with QA before the study continues. Where a run fails for a reason attributable to the equipment or to the method rather than to the parameter under study, the run may be repeated once, and both the original and the repeat will be reported in PCR-010.

7.4 Classification

Each parameter will be classified under SOP-4001 from the outcome of these studies (International Council for Harmonisation 2009, 2012). A parameter whose variation over the characterized range affects no quality attribute, but whose control is needed for the step to perform consistently, is classified as a key process parameter. A parameter whose variation affects a quality attribute is classified as critical, or as well controlled critical where the risk of falling outside the design space is low. The classification is an outcome of the studies and is reported in PCR-010.

8 Proven acceptable ranges (planned analysis)

A PAR will be assigned to each parameter from its univariate series. The acceptance basis is the drug substance criteria of Table 4.1, for the reason given in §7.2. The PAR of a parameter is the contiguous span of levels over which every attribute measured on the final retentate met its criterion, with the other parameters at their set-points. Each PAR will be reported in PCR-010 together with the measured series and the criterion against which it was read.

Three bounds apply to a PAR determined this way, and PCR-010 will state them where it states the ranges.

First, a univariate PAR holds with the other parameters at their set-points. It does not describe the step when two parameters sit at their range edges at the same time. Second, no Monte Carlo propagation of the other parameters within their normal operating ranges will be performed. That analysis predicts a response at settings the design did not run, which requires a fitted multivariate model, and no such model exists for this step. Third, a PAR cannot be wider than the range studied. Where every level in a range is acceptable, the claimed PAR is the characterization range of Table 6.1 and nothing beyond it.

The ranges reported under this analysis become the basis of the operating ranges the receiving site runs to, together with the classification of §7.4. Where a PAR is wider than the NOR, the margin between them is what a deviation is dispositioned against. Where a PAR is truncated at an edge of failure, the NOR should be reviewed against the truncated range before the step is transferred.

9 Data management and integrity

Data generated under this plan will be recorded so that they are attributable, legible, contemporaneous, original and accurate, and so that they remain complete, consistent, enduring and available. Run records will be captured in the electronic notebook under the study number. Analytical results will be captured in the laboratory information management system and reported against the sample identifiers of Table 5.1.

The statistical analysis of §5.5 will be scripted and held under version control, so that it can be re-executed on the retained raw data and return the same result. Every calculation will be reviewed by a second qualified person. A change to a record after it has been made carries an audit trail with the reason for the change. Values quoted in PCR-010 will be read from the retained data set rather than re-entered by hand. Records will be retained under the sponsor's record retention procedure for the life of the product.

10 Roles and responsibilities

The functions responsible for the study and their responsibilities are listed in Table 10.1. Process Development owns the study. Quality Assurance approves this plan before any run is executed.

Table 10.1: Functions responsible for the study.

Function	Responsibility
Process Development	Designs the study, executes the runs, analyses the data and drafts PCR-010
Manufacturing Science and Technology	Owns the scale-down model and its qualification, and carries the outcome into the transfer
Analytical Development	Releases the methods, runs the assays and reports method variability
Statistics	Reviews the design before execution and the analysis before report approval
Quality Assurance	Approves this plan and PCR-010, and dispositions any deviation from it
Receiving site process science	Reviews the resulting ranges against the commercial skid and confirms they are achievable

The receiving site is represented in the review of this plan and in the review of PCR-010, so that the ranges the studies support are assessed against the commercial skid before they are written into the control strategy (International Council for Harmonisation 2008).

11 Deliverables and schedule

The deliverable of this plan is PCR-010, the process characterization report for the step. PCR-010 reports the qualification of the scale-down model, the three univariate series, the buffer exchange curve, the PARs and the parameter classification, and it rolls up into PCMR-001.

The work runs in five phases. The scale-down model is qualified first, and it is accepted against §7.1 before any characterization run starts. The characterization runs of §6.2 are executed next, from a single load pool. The analysis of §5.5 and the PAR determination of §8 follow the last run. PCR-010 is then drafted, reviewed and approved. The classification and the ranges are finally carried into the control strategy and into PCMR-001. No calendar dates are given here. The campaign schedule is held in PCMP-001 and the transfer milestones in PTP-001.

12 Risks and assumptions

The scale-down model gives more passes through the pump per gram of antibody than the commercial skid, as stated in §5.1. The model is therefore conservative for aggregate formation, and a result that meets the criterion at bench scale is expected to hold at commercial scale. If aggregate fails its criterion at bench scale, the condition will be repeated at pilot scale before an operating range is narrowed on the strength of the bench result.

The study depends on a single Step 9 pool for its load. If one pool is not sufficient for the whole study, a second pool will be bridged by running the set-point condition on both pools and comparing the responses against the pure error band of §5.5. A difference larger than that band will be investigated before the second pool is used, and the runs executed on each pool will be identified in PCR-010.

Aggregate levels in A-Mab drug substance sit well inside the criterion, and the measured change across the step may be small against the variability of the SEC method. Noise near the low end of the method range will be attributed to the assay rather than to the parameter under study, unless the change measured across the range is larger than the pure error band of §5.5.

Three assumptions underlie the plan. The composition of the formulation buffer is fixed by the drug product programme. A change to the formulation would change the target of the buffer exchange and would invalidate the diavolume study. The membrane chemistry and the membrane supplier are as qualified, and a change to either requires a new risk assessment before these data are applied. The material entering the step arrives within the in-process limits of Step 9. The study characterizes the step under representative load, and it does not challenge the step with material that is already outside those limits.

13 Approvals

This plan takes effect when it has been approved by the functions recorded in Table 1. A change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan, approved by the same functions, before the affected runs are executed. A departure from the plan during execution will be recorded as a deviation, dispositioned by QA, and reported with its impact assessment in PCR-010.

14 Appendix A. Planned univariate level schedule

Table 14.1 lists the levels at which each parameter will be run. The low and high levels are the edges of the characterization range and the set-point level is common to all three series. No coded design matrix is given for this step, because the study carries no factorial design. The design matrices of the multivariate steps are held in their own plans.

Table 14.1: Planned levels for the univariate studies.

Parameter	Unit	Low level	Set-point	High level	NOR	Study type
Number of diavolumes	DV	7	8	10	7 to 9	univariate
Transmembrane pressure	psi	10	12	15	10 to 14	univariate
Final DS concentration	g/L	65	75	85	70 to 80	univariate

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.

International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.

International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.